3.8 G: Differentiation

	Content
G1	Understand and use the derivative of $f(x)$ as the gradient of the tangent to the graph of $y = f(x)$ at a general point (x , y); the gradient of the tangent as a limit; interpretation as a rate of change; sketching the gradient function for a given curve; second derivatives; differentiation from first principles for small positive integer powers of x and for sin x and cos x Understand and use the second derivative as the rate of change of gradient; connection to convex and concave sections of curves and points of inflection.

	Content
G2	Differentiate x^n , for rational values of n , and related constant multiples, sums and differences. Differentiate e^{kx} and a^{kx} , sin kx , cos kx , tan kx and related sums, differences and constant multiples. Understand and use the derivative of ln x

	Content
G3	Apply differentiation to find gradients, tangents and normals, maxima and minima and stationary points, points of inflection. Identify where functions are increasing or decreasing.

	Content
G4	Differentiate using the product rule, the quotient rule and the chain rule, including problems involving connected rates of change and inverse functions.

	Content
G6	Construct simple differential equations in pure mathematics and in context, (contexts may include kinematics, population growth and modelling the relationship between price and demand).